- Finally, after typing "ij", the complete input "abcdefghij" is displayed.

This output demonstrates how the debouncing effect delays the display of the input until the user pauses, providing a more efficient and controlled output.

For this test, are you able to answer these follow-up/probing questions?

- "What would happen if you did not clean up the timeout in your useDebounce hook?"
- "How would you test the useDebounce hook to ensure it behaves correctly with rapid input changes?"
- "Can you modify your `useDebounce` hook to support leading-edge or trailing-edge invocation?"
- "Why would you choose a custom hook over using an existing debounce utility like lodash’s debounce?"
- "Could your hook introduce memory leaks? Why or why not?"

4.4 React component lifecycle and patterns

Component lifecycle, controlled components, suspense & concurrency, and error handling are essential React topics frequently explored in technical interviews. Jane used table 4.9 to systematically review foundational concepts, applied skills, and advanced practices for mastering these areas.

Table 4.9 React component lifecycle and patterns

Topic	Fundamentals	Applied skills	Advanced
Component lifecycle	Describe the lifecycle phases of Mounting, Updating, and Unmounting. Class and functional component lifecycles are	Debug common issues with `useEffect` cleanup.	Optimize performance during lifecycle transitions in complex applications.